UNDERSTAND AI

Prediction

You just watched a token's final vector land in its neighborhood on the map. In Vector Space, that landing was about meaning, the model understanding the words you put in. Prediction is the flip side, the output: it takes the position at the end of your text and turns it into the next word.

THE RANKED LIST

The model doesn't just grab the single closest token. It scores **every** possible next token at once and lines them up by probability: a ranked list, with the best fit on top and a long tail of less likely options below it.

Then it takes one, almost always from the top, adds it to the text, and runs the whole thing again for the token after that. That pick has a name: **prediction**. Score every possible next token, choose one, then repeat.

ONE DIAL ON THE PICK: TEMPERATURE

There's a control on how boldly the model commits to the top of that ranked list: **temperature**.

Temperature is the variation dial. It changes how predictable or surprising the model is when choosing words. Low temperature makes answers more repeatable. High temperature makes answers more varied, creative, and sometimes weird.

Temperature does not make the model smarter. It does not make an answer true. It changes how much the model sticks to its most likely next choice.

❄️ **Low temperature**
Repeatable, predictable, less varied.

⚖️ **Medium temperature**
Balanced, natural. Usually best for normal writing and brainstorming.

🔥 **High temperature**
Surprising, unusual, sometimes weird or incoherent.

How temperature reshapes the picks

Here's that ranked list for "See you ___". Watch how the same candidates redistribute as temperature rises. This is a simplified demo: real models use other sampling controls too, and exact numbers vary by model.

	❄️ Low	⚖️ Medium	🔥 High
tomorrow	100%	65%	40%
later	0%	20%	28%
soon	0%	10%	17%
there	0%	3%	7%
again	0%	1%	4%
other words	0%	1%	4%

You probably won't see this slider in regular ChatGPT, Claude, or Gemini. The app usually picks the temperature for you. But the idea still matters.

WHEN YOU WANT REPEATABILITY VS VARIETY

Most students won't touch a temperature slider directly. But you can ask for the effect through prompting.

❄️ **When you want repeatability**

Ask for "the standard answer," "the most likely interpretation," or "your best single answer." Good for formatting, extracting information, following a template, or getting a stable result you can rely on.

🔥 **When you want variety**

Ask for "ten different options," "surprise me," or "give me unusual angles." Good for naming, story ideas, jokes, alternate phrasings, or creative exploration. Regenerate also helps.

WHAT TEMPERATURE IS NOT

Low temperature doesn't mean correct. It means repeatable. A low-temp model will give you the same answer every time, even when that answer is wrong. Temperature controls variation, not accuracy.

These odds didn't come from nowhere. The model learned every one of them from billions of examples, long before you typed a word. That's **training**.